

Logical Operators - Simple Hinglish Summary

Source: <https://javascript.info/logical-operators>

Main baat

JavaScript me **4 logical operators** hain: `(||)` (OR), `(&&)` (AND), `(!)` (NOT), `(??)` (Nullish Coalescing). Is chapter me pehle 3 hain, `(??)` agle chapter me hai.

Inhe "logical" kehte hain, lekin ye **kisi bhi type ki value** par lag sakte hain aur result bhi **kisi bhi type ka** ho sakta hai.

1. `(||)` (OR)

```
javascript  
  
result = a || b;
```

Classical boolean logic: Koi ek bhi `(true)` ho to `(true)`, dono `(false)` ho tabhi `(false)`.

```
javascript  
  
alert( true || true ); // true  
alert( false || true ); // true  
alert( true || false ); // true  
alert( false || false ); // false
```

Non-boolean operand ho to boolean me convert hota hai (`(1) → (true)`, `(0) → (false)`).

Zyadatar `(if)` me use hota hai:

```
javascript  
  
let hour = 12;  
let isWeekend = true;  
  
if (hour < 10 || hour > 18 || isWeekend) {  
    alert( 'The office is closed.' );  
}
```

2. OR pehli truthy value dhundhta hai

JavaScript me `(||)` classical se zyada powerful hai.

```
javascript
```

```
result = value1 || value2 || value3;
```

Kaise kaam karta hai:

1. Operands ko **left se right** evaluate karta hai.
2. Har operand ko boolean me convert karke dekhta hai. `(true)` mile to **ruk jata hai** aur us operand ki **original value** return karta hai.
3. Sab `(false)` hon to **aakhri operand** return karta hai.

Yaani: OR pehli truthy value return karta hai, ya aakhri value agar koi truthy na mile.

javascript

```
alert( 1 || 0 );           // 1
alert( null || 1 );        // 1
alert( null || 0 || 1 );   // 1
alert( undefined || null || 0 ); // 0 (sab falsy, aakhri wali)
```

Iske do fayde

1. List me se pehli truthy value lena:

javascript

```
let firstName = "";
let lastName = "";
let nickName = "SuperCoder";

alert( firstName || lastName || nickName || "Anonymous" ); // SuperCoder
```

Sab falsy hote to `"Anonymous"` dikhta.

2. Short-circuit evaluation: `(||)` pehli truthy value milte hi **ruk jata hai**, baaki operands ko chhuta bhi nahi. Ye tab important hai jab operand me side effect ho (function call, assignment).

javascript

```
true || alert("not printed"); // alert nahi chalega
false || alert("printed");    // alert chalega
```

3. `(ampersand)` (AND)

javascript

```
result = a && b;
```

Classical logic: Dono truthy ho tabhi `true`, warna `false`.

javascript

```
alert( true && true );    // true
alert( false && true );   // false
alert( true && false );   // false
alert( false && false );  // false
```

javascript

```
let hour = 12;
let minute = 30;

if (hour == 12 && minute == 30) {
  alert( 'The time is 12:30' );
}
```

4. AND pehli falsy value dhundhta hai

javascript

```
result = value1 && value2 && value3;
```

Kaise kaam karta hai:

1. Left se right evaluate karta hai.
2. Har operand ko boolean me convert karta hai. `false` mile to **ruk jata hai** aur us operand ki **original value** return karta hai.
3. Sab truthy hon to **aakhri value** return karta hai.

Yaani: AND pehli falsy value return karta hai, ya aakhri value agar sab truthy hon.

OR aur AND ka fark: OR pehli truthy dhundhta hai, AND pehli falsy dhundhta hai.

javascript

```

alert( 1 && 0 );           // 0
alert( 1 && 5 );           // 5
alert( null && 5 );        // null
alert( 0 && "no matter" ); // 0

alert( 1 && 2 && null && 3 ); // null (pehli falsy)
alert( 1 && 2 && 3 );        // 3 (sab truthy, aakhri)

```

Precedence: `&&` ki precedence `||` se **zyada** hai. Yaani `(a && b || c && d) = (a && b) || (c && d)`.

`if` ki jagah `||` ya `&&` mat use karo

javascript

```

let x = 1;

(x > 0) && alert( 'Greater than zero!' ); // chalta hai
if (x > 0) alert( 'Greater than zero!' ); // behtar, zyada clear

```

Dono same kaam karte hain, lekin `if` zyada readable hai. **Jis kaam ke liye jo bana hai, wahi use karo.**

5. `!` (NOT)

javascript

```
result = !value;
```

1. Operand ko boolean me convert karta hai.
2. **Ulta value** return karta hai.

javascript

```

alert( !true ); // false
alert( !0 );    // true

```

Double NOT `!!`

Value ko **boolean me convert** karne ke liye use hota hai:

javascript

```
alert( !!"non-empty string" ); // true
alert( !!null );                // false
```

Pehla (!) convert karke ulta karta hai, dusra (!) wapas ulta kar deta hai. Ye Boolean(...) jaisa hi hai:

javascript

```
alert( Boolean("non-empty string") ); // true
alert( Boolean(null) );                // false
```

Precedence: (!) ki precedence **sabse zyada** hai, ye (&&) aur (||) se pehle chalta hai.

Practice Tasks (Answers)

1. `alert( null || 2 || undefined );` → (2) (pehli truthy)
2. `alert( alert(1) || 2 || alert(3) );` → pehle (1), phir (2) `alert( undefined )` return karta hai (falsy), isliye OR aage badhta hai. (2) truthy hai, wahin ruk jata hai, `alert(3)` chalta hi nahi.
3. `alert( 1 && null && 2 );` → `null` (pehli falsy)
4. `alert( alert(1) && alert(2) );` → pehle (1), phir `undefined` `alert(1)` `undefined` (falsy) return karta hai, AND wahin ruk jata hai.
5. `alert( null || 2 && 3 || 4 );` → (3) (&&) pehle chalta hai: `2 && 3 = 3`. Phir `null || 3 || 4 = 3`.

6. Age 14 se 90 ke beech (inclusive):

javascript

```
if (age >= 14 && age <= 90)
```

7. Age 14 se 90 ke beech NAHI hai:

javascript

```
// NOT ke saath
if (!(age >= 14 && age <= 90))

// NOT ke bina
if (age < 14 || age > 90)
```

8. Kaun se alert challenge?

javascript

```
if (-1 || 0) alert( 'first' );           // chalega (-1 truthy)
if (-1 && 0) alert( 'second' );          // nahi (0 falsy)
if (null || -1 && 1) alert( 'third' );   // chalega (-1 && 1 = 1)
```

Jawab: **pehla aur teesra.**

9. Login check (nested if):

javascript

```
let userName = prompt("Who's there?", '');

if (userName === 'Admin') {

    let pass = prompt('Password?', '');

    if (pass === 'TheMaster') {
        alert( 'Welcome!' );
    } else if (pass === '' || pass === null) {
        alert( 'Canceled' );
    } else {
        alert( 'Wrong password' );
    }

} else if (userName === '' || userName === null) {
    alert( 'Canceled' );
} else {
    alert( "I don't know you" );
}
```

(Khali input `''` deta hai aur `Esc` `null` deta hai.)

Quick Summary

Operator	Kya karta hai	Return karta hai
<code> </code> (OR)	Pehli truthy dhundhta hai	Pehli truthy, ya aakhri value
<code>&&</code> (AND)	Pehli falsy dhundhta hai	Pehli falsy, ya aakhri value
<code>!</code> (NOT)	Value ulti karta hai	<code>true</code> / <code>false</code>
<code>!!</code>	Boolean me convert karta hai	<code>true</code> / <code>false</code>

Precedence: `!` > `&&` > `||`

Yaad rakho:

- `||` aur `&&` **original value** return karte hain, sirf `true/false` nahi.
- Dono **short-circuit** karte hain (pehla result mil gaya to baaki chhod dete hain).
- `if` ki jagah `||` / `&&` mat use karo.